Lecture plan · for the lecturer

Loss, descent and backprop

University, 2

Time
40 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Loss is large when wrong
- Step against the slope
- Backprop: each weight's share of the blame

Activity: Steps by hand

1. Predict (5 min): Pairs predict which of three learning rates will climb out of the valley of θ^2, and write why.
2. By hand (10 min): Each pair takes five steps on θ^2 from $\theta = 3$ with $\eta = 0.1$, 0.5 and 1.5. Mark the arithmetic, and ask where the gradient pointed.
3. Backward (10 min): Work $y = w \times x$ with the loss $(y - t)^2$ on the board. Pairs find the gradient for w with the chain rule and say which way w moves.
4. Lab (10 min): One phone between two: run the Gradient descent lab on Good, then Too big, and check the predictions. Collect one wrong prediction and why it was tempting.
5. Exit (5 min): The exit ticket: θ^2 at 3, rate 0.1, write the next θ.

Discussion

- Where does the gradient point?

Exit ticket

- θ^2 at 3, rate 0.1: write the next θ.

Try it in your next lecture
Bring three loss curves to the seminar with the rates removed. Students name the rate.

Seminar sheet · for students

Loss, descent and backprop

University, 2

Name _____ Group _____ Date _____

1 **On the loss θ^2, a student's steps take θ from 3 to 6 to 12. What went wrong?**

◯ Each step went with the gradient, uphill ◯ The rate is too big ◯ The loss has no minimum

2 **The gradient points at the minimum.**

◯ True ◯ False

3 **Match the curve to the cause.**

Loss grows every step	• • • •	Step too small
Loss falls very slowly	• • • •	Memorising the training data
Train loss low, new data worse	• • • •	Step too big

4 **Order one descent step.**

☐ Compute the gradient

☐ Step against the gradient

☐ Compute the loss

5 **One weight: $y = w \times x$, and the loss is $(y - t)^2$. With $x = 2$, $w = 1$ and $t = 3$, what is the gradient of the loss with respect to w?**

◯ 4 ◯ -2 ◯ -4

Answers · for the lecturer

Loss, descent and backprop

University, 2

1 **Each step went with the gradient, uphill**

A rate that is too big flips the sign of θ at every step (3, −6, 12).

2 **False**

It points uphill, locally. Descent steps against it.

3 **Loss grows every step — Step too big**
Loss falls very slowly — Step too small
Train loss low, new data worse — Memorising the training data

The learning rate and overfitting leave different fingerprints on the curve.

4 **Top to bottom: 2, 3, 1**

You need the slope before you can step against it.

5 **−4**

The derivative of the loss with respect to y is 2(y − t) = −2, and the derivative of y with respect to w is x = 2.

For you to think about

What will you mark if the arithmetic is right and the sentence says the gradient points at the minimum?